

Principal Curvatures κ_1 κ_2 along the principal directions forming the lines of curvature for a function $h(u,v)$ in Cartesian coordinates.

$$\kappa_1 = \frac{h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1 + h_v^2 + h_u^2)^{\frac{3}{2}}} - \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1 + h_v^2 + h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1 + h_v^2 + h_u^2)^2}}$$

$$\kappa_2 = \frac{h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1 + h_v^2 + h_u^2)^{\frac{3}{2}}} + \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1 + h_v^2 + h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1 + h_v^2 + h_u^2)^2}}$$

Note that κ_1 $\kappa_2 > 0$ if the Hessian is positive definite; ie $h_{uu} h_{vv} - h_{uv}^2 > 0$

Monge/Local Astigmatism $\kappa_1 - \kappa_2$

$$A = 2 \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1 + h_v^2 + h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1 + h_v^2 + h_u^2)^2}}$$

Mean curvature $(\kappa_1 + \kappa_2)/2$

$$H = \frac{h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1 + h_v^2 + h_u^2)^{\frac{3}{2}}}$$

Gaussian Curvature $K = \kappa_1 \kappa_2$

$$K = \frac{h_{vv} h_{uu} - h_{uv}^2}{(1 + h_v^2 + h_u^2)^2}$$

The principal curvatures are also expressible as

$$\kappa_1 \kappa_2 = H \pm \sqrt{H^2 - K} = H \pm A/2$$

In polar coordinates this gets more cumbersome, though the ubiquitous denominator of first derivatives is quite simple, so for $r > 0$,

$$h_u = h_r \cos(t) - \sin(t) \frac{h_t}{r}$$

$$h_v = h_r \sin(t) + \cos(t) \frac{h_t}{r}$$

$$(1 + h_v^2 + h_u^2) = \left(1 + h_r^2 + \left(\frac{h_t}{r}\right)^2\right)$$

The “lines of curvature” describing the directions of the principal curvatures can be plotted as a vector field in (u,v)

The principal directions can be given by

$$t_1 = \begin{bmatrix} h_{uv} \\ \kappa_1(1 + h_x^2 + h_y^2) - h_{uu} \end{bmatrix}$$

$$t_2 = \begin{bmatrix} \kappa_2(1 + h_x^2 + h_y^2) - h_{vv} \\ h_{uv} \end{bmatrix}$$

(these last two equations come from Eric Albin, Ronnie Knikker, Shihe Xin, Christian Oliver Paschereit, Yves d’Angelo. Computational assessment of curvatures and principal directions of implicit surfaces from 3D scalar data. Lecture Notes in Computer Science, 2017, Mathematical Methods for Curves and Surfaces, 10521, pp.1-22. {10.1007/978-3-319-67885-6_1}. {hal-01486547})

Care must be taken with the signs.

The conversion for the second derivatives (Hessian) is

$$h_{uu} = h_{rr} - \sin^2(t) \left(h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) + 2 \cos(t) \sin(t) \left(\frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

$$h_{vv} = h_{rr} - \cos^2(t) \left(h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) - 2 \cos(t) \sin(t) \left(\frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

$$h_{uv} = \cos(t) \sin(t) \left(h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) + (\sin^2(t) - \cos^2(t)) \left(\frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

Curiously, if at the origin with $h_t \neq 0$ then as $r \rightarrow 0$, $K \rightarrow \left(\frac{h_{tt}}{h_t} \right)^2$

Of course the ratio of h_t of to h_{tt} cannot be a function of t at $r=0$, because that would imply that K is not single valued at the origin. One way to achieve independence from t is if the surface is axisymmetric, then at the origin:

$K = (h_{rr})^2$, $H = h_{rr}$ as follows from the axisymmetric versions of the equations below.

Simplified formulae for surfaces of revolution (axisymmetric surfaces with elevation $z(r)$ where r is the radius from the axis of revolution) and all derivatives in θ are zero:

$$z_\theta = 0, \quad z_{r\theta} = 0, \quad \text{and} \quad z_{\theta\theta} = 0$$

"Instantaneous or Tangential" curvature, polar coordinates:

$$\kappa_1 = \frac{z_{rr}}{(1+z_r^2)^{3/2}}$$

which actually the formula for the first principal curvature for a surface of revolution (axisymmetric) parameterized by $z=z(r)$. The other principal curvature in that case is actually the

"Axial or Sagittal" curvature, polar coordinates:

$$\kappa_2 = \frac{z_r}{r\sqrt{1+z_r^2}}, \text{ with } z_r \rightarrow 0 \text{ at } r \rightarrow 0 \text{ implies } \kappa_2 \rightarrow z_{rr}$$

Note also that the tangential curvature is the derivative of the sagittal curvature in this case.

As an example, consider a simple paraboloid, $z(r)=r^2$. The ratio between $\kappa_1/\kappa_2 = (1+4r^2)$, showing that for $r \ll 1$, the two are similar, but as the radius increases, the axial curvature becomes larger than the tangential curvature. Since the two principal curvatures are orthogonal, that also allows us to realize that the axial curvature is in the radial direction, whereas the tangential curvature is in the circumferential direction.

Mean Power is the arithmetic average $(\kappa_1 + \kappa_2)/2$. H is traditionally used.

$$H = \frac{1}{2} \left[\frac{z_r}{r\sqrt{1+z_r^2}} + \frac{z_{rr}}{(1+z_r^2)^{3/2}} \right]$$

Gaussian Power is the product $\kappa_1\kappa_2$ of the principal curvatures. It's unfortunate that it isn't the geometric mean instead of its square...

$$K = \frac{z_r z_{rr}}{r(1+z_r^2)}$$

So in order to get a comparable "power" in diopters, the square root of K is used.

So called Monge Astigmatism $\kappa_1 - \kappa_2$:

$$A = \left[\frac{z_r}{r \sqrt{1+z_r^2}} - \frac{z_{rr}}{(1+z_r^2)^{3/2}} \right]$$

Warp/Oblique power, same as axial but with z_θ/r instead of z_r
This last definition is non-standard.

$$K = \frac{z_\theta}{r^2 \sqrt{1+(z_\theta/r)^2}},$$

The principal directions in the axisymmetric case can be given by the meridians and circles of longitude, as can easily be seen by noting that in the equations given above $h_{uv} = 0$, giving principal directions in either the r or circumferential (angular or θ) direction.